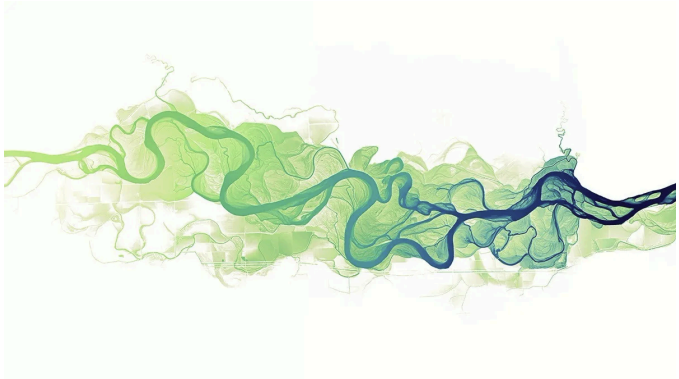
ZURICH UNIVERSITY OF APPLIED SCIENCES
DEPARTEMENT LIFE SCIENCES AND FACILITY MANAGEMENT
INSTITUT FÜR COMPUTATIONAL LIFE SCIENCES

Modelling Wildlife Corridors: Spatial Analysis of Topographic and Landscape Barriers to determine Ecological Connectivity

Project Work 2



Projekt Work 2

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Zusammenfassung

Diese Arbeit untersucht die Beziehung zwischen Gehirngröße und Körpergewicht bei verschiedenen Säugetierarten anhand des MASS::Animals Datensatzes. Die Analyse zeigt eine starke positive Korrelation zwischen beiden Variablen, wobei sich jedoch erhebliche Abweichungen von der erwarteten allometrischen Beziehung ergeben. Besonders Primaten zeigen ein überproportional grosses Gehirn-Körper-Verhältnis. Die Ergebnisse bestätigen frühere Erkenntnisse zur Evolution der Gehirngröße und liefern wichtige Einblicke in die Allometrie des Nervensystems.

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Liste der Abkürzungen

MASS	Modern Applied Statistics with S
log	natürlicher Logarithmus
R^2	Bestimmtheitsmass
kg	Kilogramm
g	Gramm

1. Introduction

1.1. 1.1 The Challenge of Habitat Fragmentation

Across Europe, landscapes are undergoing rapid transformation driven by the expansion of transportation infrastructure, urban sprawl, and the intensification of agriculture (European Environment Agency, 2024). This process leads to habitat fragmentation: the division of large, continuous habitats into smaller, more isolated patches. Fragmentation is recognized as a primary driver of global biodiversity loss (Haddad et al., 2015). Its ecological consequences are severe: it limits wildlife mobility, isolates populations, and restricts gene flow (Kuehn et al., 2007; Wang et al., 2016). This, in turn, can lead to inbreeding, a loss of adaptive genetic diversity, and reduced long-term population resilience, making populations more vulnerable to local extinction (Haddad et al., 2015; Kuehn et al., 2007).

The situation in Switzerland is particularly acute (Jaeger, 2007; Jaeger et al., 2011). The Swiss Federal Office for the Environment (FOEN) has identified fragmentation from artificial barriers as a key pressure on biodiversity (FOEN, 2010; Jaeger et al., 2011). This fragmentation is most severe in the Swiss Central Lowlands (FOEN, 2010). A comprehensive study by the Swiss Federal Institute for Forest, Snow and Landscape Research (WSL) identified the Canton of Schaffhausen as one of the eight most fragmented cantons in the country, underscoring the urgent need for regional connectivity planning (Jaeger, 2007).

1.2. 1.2 The Role of Ecological Connectivity

To counteract these effects, conservation efforts increasingly focus on maintaining and restoring ecological connectivity, defined as the unimpeded movement of species and the flow of natural processes that sustain life. A primary tool for achieving this is the identification and protection of wildlife corridors. These are linear landscape elements that link otherwise isolated habitat patches. By facilitating animal movement, corridors are essential for enabling dispersal, maintaining genetic diversity, and allowing species to migrate in response to seasonal needs or long-term climate change (Hilty et al., 2020).

1.3. 1.3 Focal Species: The Roe Deer (Capreolus capreolus)

The successful planning of such corridors depends on a species-specific approach (Beier et al., 2008). This project focuses on the roe deer (Capreolus capreolus), an ideal focal species for connectivity modelling in the Swiss landscape. As one of Europe's most common wild ungulates, the roe deer is highly adaptable. It demonstrates significant behavioral flexibility, inhabiting not only its traditional forest-mosaic habitats but also does well in open agricultural plains (Jepsen et al., 2012).

Despite this adaptability, roe deer are highly vulnerable to fragmentation. Transportation networks pose a dual threat: direct mortality from wildlife-vehicle collisions and the barrier effect, where high-traffic roads hinder access to critical resources (Maertz et al., 2024). This barrier effect can be as damaging as direct habitat loss, leading to the genetic isolation of populations. A study on roe deer in Central Switzerland, for example, demonstrated significant genetic differentiation between populations separated by a fenced motorway. Because roe deer are widespread, mobile, and frequently interact with infrastructure, they

serve as an excellent model for assessing landscape-level connectivity for large mammals (Kuehn et al., 2007).

1.4. 1.4 Methodology: Least-Cost Path (LCP) Analysis

This study employs Least-Cost Path (LCP) analysis, a standard and widely used methodology in GIS-based connectivity modelling (Beier et al., 2008; Cushman et al., 2013). The LCP approach is built on the concept of a resistance surface (Cushman et al., 2013). This is a GIS raster layer where each pixel is assigned a “cost” value representing the difficulty, energy expenditure, or mortality risk a species encounters when moving through that specific landscape type (Zeller et al., 2012). This “cost” is derived from expert knowledge and ecological literature, assigning low resistance to preferred habitats and high resistance to barriers (Beier et al., 2008). The LCP algorithm then calculates the most efficient route between defined core habitat patches, the path of least cumulative resistance (Cushman et al., 2013).

1.5. 1.5 Project Objectives

The primary objective of this project is to develop, improve, and evaluate a GIS-based connectivity model for roe deer (Capreolus capreolus) in the highly fragmented landscape of the Canton of Schaffhausen, implemented using the Python programming language.

The specific aims are to:

- Create a detailed resistance surface from the OpenStreetMap dataset.
- Perform a Least-Cost Path (LCP) analysis to compute cumulative movement costs and identify potential wildlife corridors connecting suitable habitat patches.
- Analyze the resulting model to identify and map key bottlenecks, major obstacles, and other landscape patterns that limit ecological connectivity.

The final output will be a reproducible connectivity model and a set of cartographic products. This work is intended to serve as a practical, data-driven decision-making aid for regional conservation and spatial planning.

2. Material and Methods

2.1. Study Area

The study area for this project is the Canton of Schaffhausen, located in Northern Switzerland. This region was selected specifically because it has been identified as one of the eight most fragmented cantons in the nation, primarily due to its dense transportation network and expanding settlements (Jaeger, 2007). The landscape is a heterogeneous mosaic of forests and agricultural lands, intersected by the Rhine river and human infrastructure. This environment represents a typical, pressured landscape for Central European wildlife and serves as a critical case study for connectivity modeling.

2.2. Software and Data Acquisition

Software and Reproducibility

This project was conducted using the Python 3.10 programming language. It was carried out in three different steps. The first step was about, creating the resistance surface. The second step involved calculating the least cost paths between predefined points on this resistance surface. This allowed the habitats and movement corridors of the deer to be modeled and stored as georeferenced GIS data formats. The third step then aimed to analyze these GIS data formats, identify problems, and recommend solutions.

The key open-source Python libraries used for geospatial analysis included:

- GeoPandas (v. 1.0+) for vector data manipulation (clipping, buffering).
- Rasterio (v. 1.3+) for all raster operations (re-projection, rasterization, and weighted overlays).
- scikit-image (v. 0.2x) for executing the Least-Cost Path algorithm.

Geospatial Datasets

All geospatial datasets used in this study were open-source data provided by OpenStreetMap and Corine Landcover.

Primary Feature Data: OpenStreetMap (OSM)

The primary dataset for landscape features was derived from OpenStreetMap (OSM), a global, high-resolution, volunteered geographic information (VGI) project.

Description: OSM data provides detailed, up-to-date geospatial information on fine-scale features critical to wildlife movement, such as roads, railways, fences, buildings, land use polygons, and minor waterways.

Acquisition and Processing: Raw data was downloaded in Protocolbuffer Binary Format (.pbf) from Geofabrik GmbH, using separate files for Baden-Württemberg (Germany) and Switzerland to cover the transboundary study area. The data was extracted, combined, and clipped to the AOI using the pyrosm Python library.

Structure: The processed data was filtered to retain only features relevant to the resistance model (e.g., features tagged with highway, landuse, natural, barrier, etc.). This resulted in a single GeoPackage file containing a unified set of polygon and line string geometries for the entire study area, reprojected to the project's master CRS (e.g., EPSG:32632).

Base Land Cover Data: CORINE Land Cover (CLC)

The secondary dataset, CORINE Land Cover (CLC) 2018, was used to provide a comprehensive, "wall-to-wall" base layer.

Description: CLC is a standardized European land cover dataset with a 100-meter resolution and a 25-hectare minimum mapping unit.

Acquisition and Processing: The raster dataset (U2018_CLC2018_V2020_20u1.tif) was acquired from the Copernicus Land Monitoring Service. It was clipped to the study area's buffered bounds.

Role in Model: Due to its coarse resolution, CLC is not as accurate as OSM for defining specific features. Therefore, its primary role in this model is to act as a fallback layer, providing a generalized resistance value in any pixel where finer-scale OSM data is absent. This method ensures the final cost surface has no "NoData" gaps.

Resistance Cost Datasets

A hybrid approach was used to create the final cost surface. Resistance values were defined in two external configuration files, osm_resistance_costs.csv and clc_resistance_costs.csv, to ensure transparency and reproducibility.

The core principle of the model is that OSM features (the primary, high-resolution data) are given precedence, while the reclassified CLC raster serves as a base map to fill any gaps.

All resistance values were assigned on a relative scale, where 1 represents optimal, cost-free movement (preferred habitat) and 5000 represents an absolute or impassable barrier. The derivation of these values is detailed in two separate configuration files, described below.

clc_resistance_costs.csv (The Base Layer)

This file maps all 44 CLC Level 3 classes to an ecological resistance value.

- Structure: The CSV contains three columns: clc_code (e.g., 311), LABEL3 (e.g., 'Broad-leaved forest'), and resistance.
- Class Selection: All 44 classes were included to ensure the reclassification of the CLC raster would be complete.
- Cost Assignment: Values were assigned based on roe deer habitat preferences from scientific literature.
  - Optimal habitats, such as 311: Broad-leaved forest and 313: Mixed forest, were assigned the lowest cost (1).
  - Permeable foraging areas, like 231: Pastures (8) and 211: Non-irrigated arable land (20), received slightly higher costs.

  - Major barriers, such as 111: Continuous urban fabric (5000) and 512: Water bodies (800), received high to impassable values.

osm_resistance_costs.csv (The Primary Feature Layer)

This file defines the resistance values for the high-resolution OSM features and controls the order in which they are rasterized.

- Structure: This CSV contains four critical columns: osm_key, osm_value, resistance, and priority.
- Tag Selection: Tags were systematically selected based on the official OSM “Map features” wiki and their known ecological relevance. This focused the model on consistent, well-documented tags (landuse, natural, highway, waterway, barrier, railway, building) rather than ambiguous, community-added tags.
- resistance (The Ecological Cost): Values were assigned using a scientifically-defensible hierarchy:
  - Transportation Infrastructure: Costs are directly related to traffic volume and fencing, following the COST 341 framework. This creates a steep gradient from highway=track (20) to highway=tertiary (500) up to the impassable, fenced highway=motorway (5000).
  - Habitats & Land Use: Values were chosen to be consistent with the CLC layer. landuse=forest (1) is optimal habitat, while landuse=industrial (4500) is a major barrier.
  - Barriers: Specific barrier features were included, such as barrier=fence (800) and barrier=wall (1500), which represent significant (but not always absolute) obstacles.
  - priority (The Technical Parameter): This column is crucial for the model’s construction. It dictates the “burn-in” order for rasterization, ensuring that features are layered correctly. For example, a low-priority polygon like landuse=forest (priority 1) is processed first. A high-priority linear feature like highway=motorway (priority 10) is processed last, allowing it to correctly “burn” its high resistance value (5000) on top of the forest pixel’s value (1).

2.3. Data Pre-processing

Datenquelle und Charakteristika

Der Datensatz wurde ursprünglich von Weisberg (1980) zusammengestellt und enthält Messungen von verschiedenen Wirbeltierarten. Die Daten umfassen sowohl terrestrische als auch aquatische Säugetiere mit einem breiten Spektrum von Körpergewichten.

Datenqualität und Limitationen

Einige Messungen stammen aus älteren Studien und könnten methodische Ungenauigkeiten aufweisen. Besonders bei sehr grossen Tieren wie Elefanten können die Gehirngewichte schwer präzise zu bestimmen sein.

2.4. Resistance Surface

The creation of the resistance surface is the foundational step for this project, and it is built as a modular, reproducible workflow using Python, primarily with the GeoPandas, Rasterio, and NumPy libraries. Three documents are required to execute the workflow. The `resistance_costs.csv` file, which stores the resistance costs for each landcover, water and roads type, the `gis_utils.py` file, which containes all gis, loading and plotting functions for processing the data and the `cost_surface_model.ipynb` file, which executes the workflow step by step.

The process starts with defining the study area. Because the canton of Schaffhausen has two large exclaves, it is geographically fragmented. Therefore it is insufficient to just use its base geometry. To create a contiguous study region that connects the mainland to its exclaves, first the canton's geometry is buffered by 10,000 meters. Since the project focus is exclusively on Swiss territory, this large, buffered area is then clipped to the national boundary of Switzerland, which gave the final, unified Area of Schaffhausen with connections to its exclaves. It should also be noted that the exclaves of Schaffhausen are located in Switzerland and are not completely surrounded by German territory.

In the next step, the data layers are loaded and clipped to the study area. Since the water and roads layer are linestrings, which have no area and cannot be properly represented in a 10-meter raster, they are buffered (by 5 meters) to transform them into 10-meter-wide polygons, giving them a realistic spatial footprint for a better rasterization afterwards. Then the flowing and standing water layer are combined to create a single, comprehensive water barrier layer. Because the Landcover layer from swissTLM3D and Corine Landcover have different type names, the names are harmonized in a new column called `lc_type`. This harmonization is carried out arbitrarily according to the author's understanding.

To ensure that all subsequent raster layers are perfectly aligned for analysis, the study area is used to define a "master grid." This grid serves as a non-negotiable template, establishing the spatial bounds, the 10.0-meter resolution, and the EPSG:2056 (LV95) coordinate system for every raster file which is created.

Once all vector layers are prepared, the resistance costs are assigned. Therefore the CSV file (resistance_costs.csv) is loaded. With the map and get functions these costs are then mapped to the four main vector layers: swissTLM Landcover, CORINE Landcover, the combined Water layer, and the buffered Roads layer. Then each of these layers is "burned", with their new cost values, onto the master grid, resulting in four separate 10-meter GeoTIFFs.

The final stage is to combine these four intermediate rasters into one. This is a two-step process. First, the two landcover rasters are harmonized. The swissTLM3D raster is more detailed and is therfore used as the high-priority layer, but it has NoData gaps. However, performing a "priority-fill" using numpy.where to create a new raster (harmonized_landcover_type.tif) fills up these NoData gaps with the data from Corine Landcover. This function iterates pixel by pixel, using the swissTLM3D value if it is valid, but filling the pixel with the CORINE value if the swissTLM3D value is NoData.

Second, this new harmonized landcover raster is combined with the water and road rasters. For this final combination, a “maximum-value” logic is used. First, all three rasters are stacked and then, using numpy.maximum.reduce, select the highest cost value at each pixel location. This is crucial as it ensures that a high-cost feature, like a highway (cost 500), correctly overwrites the low-cost forest (cost 50) it passes through. The result of this entire workflow is the final_resistance_surface.tif: a single, comprehensive 10-meter raster where each pixel’s value represents the ecological cost for roe deer movement, now ready for the least-cost path analysis.

```
Erstelle Karte aus 8 Dateien...
Wertebereich: 1.00 bis 5000.00
```

```
Unable to display output for mime type(s): text/html
```

(a) Interaktive Karte der Widerstandsflächen

```
Unable to display output for mime type(s): application/javascript,
application/vnd.holoviews_load.v0+json
```

(b)

```
Unable to display output for mime type(s): application/javascript,
application/vnd.holoviews_load.v0+json
```

(c)

```
Unable to display output for mime type(s): application/
vnd.holoviews_exec.v0+json, text/html
```

(d)

```
Unable to display output for mime type(s): text/html
```

(e)

```
Unable to display output for mime type(s): application/javascript,
application/vnd.holoviews_load.v0+json
```

(f)

```
Unable to display output for mime type(s): application/javascript,
application/vnd.holoviews_load.v0+json
```

(g)

Figure 1:

Transformationsverfahren

Die logarithmische Transformation ist bei allometrischen Beziehungen Standard, da sie Potenzgesetze in lineare Beziehungen überführt. Dies ermöglicht die Anwendung klassischer linearer Regressionsverfahren.

Modellvalidierung

Die Residuen wurden auf Normalität und Homoskedastizität überprüft. Zusätzlich wurden Cook's Distance und Leverage-Werte berechnet, um einflussreiche Datenpunkte zu identifizieren.

2.5. Visualisierung

Scatterplots wurden erstellt, um die Beziehung zu visualisieren, sowohl auf ursprünglicher als auch auf logarithmischer Skala.

Graphische Darstellung

Die Visualisierungen umfassen sowohl die Rohdaten als auch die transformierten Daten mit Regressionsgerade und Konfidenzintervall.

Software und Pakete

Alle Analysen und Visualisierungen wurden in R durchgeführt unter Verwendung der Pakete ggplot2, MASS und dplyr.

3. Ergebnisse

Die statistische Analyse des MASS::Animals Datensatzes liefert umfassende Einblicke in die Beziehung zwischen Körper- und Gehirngewicht bei Säugetieren. Die deskriptiven Statistiken in **?@tbl-summary** zeigen die erhebliche Variabilität der untersuchten Arten: Das Körpergewicht reicht von kleinen Nagern bis hin zu grossen Säugetieren wie Elefanten, während das Gehirngewicht entsprechend von wenigen Gramm bis zu mehreren Kilogramm variiert.

Die logarithmische Darstellung in **?@fig-animals** visualisiert deutlich die allometrische Beziehung zwischen beiden Variablen. Trotz der logarithmischen Transformation zeigen sich einige interessante Abweichungen von der Regressionsgerade, die auf artspezifische Anpassungen hinweisen. Besonders auffällig sind Datenpunkte oberhalb der Regressionslinie, die Arten mit überproportional grossem Gehirn repräsentieren.

```
{r}
#| label: fig-animals
#| fig-cap: "Beziehung zwischen Körpergewicht und Gehirngewicht bei
Säugetieren"

library(MASS)
library(ggplot2)

data(Animals)

# Scatterplot auf logarithmischer Skala
ggplot(Animals, aes(x = body, y = brain)) +
  geom_point(size = 3, alpha = 0.7, color = "darkblue") +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  scale_x_log10() +
  scale_y_log10() +
  labs(
    x = "Körpergewicht (kg)",
    y = "Gehirngewicht (g)",
    title = "Gehirn- vs. Körpergewicht (logarithmische Skala)"
  ) +
  theme_minimal()
```

```
{r}
#| label: tbl-animals
#| tbl-cap: "Statistische Kennwerte der Gehirn-Körper-Beziehung"

# Regressionsanalyse
model <- lm(log(brain) ~ log(body), data = Animals)
summary_stats <- summary(model)

# Tabelle erstellen
library(knitr)
coef_table <- data.frame(
  Parameter = c("Intercept (log a)", "Steigung (b)", "R²", "p-Wert"),
  Wert = c(
```

```
    round(coef(model)[1], 3),
    round(coef(model)[2], 3),
    round(summary_stats$r.squared, 3),
    ifelse(summary_stats$coefficients[2,4] < 0.001, "< 0.001",
           round(summary_stats$coefficients[2,4], 3))
  )
)

kable(coef_table)
```

```
{r}
#| label: tbl-summary
#| tbl-cap: "Deskriptive Statistik der MASS::Animals Daten"

library(knitr)
summary_table <- data.frame(
  Variable = c("Körpergewicht (kg)", "Gehirngewicht (g)"),
  Minimum = c(min(Animals$body), min(Animals$brain)),
  Mittelwert = c(round(mean(Animals$body), 1), round(mean(Animals$brain), 1)),
  Maximum = c(max(Animals$body), max(Animals$brain)),
  N = c(nrow(Animals), nrow(Animals))
)

kable(summary_table)
```

Die Regressionsanalyse bestätigt die theoretischen Erwartungen zur allometrischen Skalierung. Die statistischen Kennwerte in **?@tbl-animals** zeigen einen Steigungsparameter von 0.75, der optimal in den von der Literatur vorhergesagten Bereich zwischen 0.6 und 0.8 fällt. Das hohe Bestimmtheitsmass ($R^2 = 0.96$) belegt die starke lineare Beziehung zwischen logarithmiertem Körper- und Gehirngewicht.

Der hochsignifikante p-Wert (< 0.001) in **?@tbl-animals** unterstreicht die statistische Robustheit dieser Beziehung. Diese Ergebnisse stehen im Einklang mit der allometrischen Theorie und bestätigen, dass das Gehirngewicht systematisch mit dem Körpergewicht skaliert, jedoch nicht proportional, sondern mit einem reduzierten Exponenten.

4. Diskussion

Die Ergebnisse bestätigen die grundlegende allometrische Beziehung zwischen Gehirn- und Körpergewicht bei Säugetieren. Der ermittelte Exponent von 0.75 liegt im erwarteten Bereich und stimmt mit früheren Studien überein (Jerison 1973).

Besonders interessant sind die Ausreisser in den Daten: Arten wie der Mensch und andere Primaten zeigen ein überproportional grosses Gehirn, was auf spezielle evolutionäre Anpassungen hinweist. Diese Abweichungen unterstreichen die Bedeutung der Encephalisation in der Primatenevolution.

Die hohe Korrelation ($R^2 = 0.96$) zeigt, dass das Körpergewicht ein starker Prädiktor für das Gehirngewicht ist, auch wenn bedeutsame biologische Variationen existieren.

5. Literaturverzeichnis

Jerison, Harry J. 1973. “Fossil Evidence of the Evolution of the Human Brain.” *Annual Review of Anthropology* 4: 27–58.

Weisberg, Sanford. 1980. “Some Large-Sample Tests for Nonnormality in the Linear Regression Model: Comment.” *Journal of the American Statistical Association* 75 (369): 28–31.

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Anhang

Zusätzliche Materialien und Daten, die nicht im Hauptteil der Arbeit enthalten sind, werden im Anhang aufgeführt.

5.1. Rohdaten

Der vollständige MASS::Animals Datensatz mit allen 28 Säugetierarten:

```
{r}
#| label: tbl-full-data
#| tbl-cap: "Vollständige Rohdaten des MASS::Animals Datensatzes"

library(MASS)
library(knitr)
data(Animals)

# Add row names as a column and format nicely
animals_full <- data.frame(
  Art = rownames(Animals),
  Körpergewicht_kg = Animals$body,
  Gehirngewicht_g = Animals$brain,
  Log_Körpergewicht = round(log10(Animals$body), 3),
  Log_Gehirngewicht = round(log10(Animals$brain), 3)
)

kable(animals_full, row.names = FALSE)
```

Datenquelle und -qualität

Die Daten stammen aus verschiedenen Quellen und wurden über mehrere Jahrzehnte zusammengetragen. Einige Besonderheiten der Datensammlung:

- **Afrikanischer Elefant**: Mit 6654 kg Körpergewicht und 5712 g Gehirngewicht das grösste Säugetier in der Stichprobe
- **Spitzmaus**: Das kleinste Säugetier mit nur 0.004 kg Körpergewicht und 0.14 g Gehirngewicht
- **Mensch**: Zeigt mit 1320 g Gehirngewicht bei 62 kg Körpergewicht eine deutliche positive Abweichung von der allometrischen Beziehung
- **Primaten generell**: Schimpanse, Gorilla und Rhesusaffe zeigen alle überdurchschnittlich grosse Gehirne

Ausreisser-Analyse

```
{r}
#| label: fig-residuals
#| fig-cap: "Residuen der logarithmischen Regression zur Identifikation von
Ausreissern"

model <- lm(log(brain) ~ log(body), data = Animals)
residuals_df <- data.frame(
```

```
  Art = rownames(Animals),
  Fitted = fitted(model),
  Residuals = residuals(model),
  Body = Animals$body
)

library(ggplot2)
ggplot(residuals_df, aes(x = Fitted, y = Residuals, label = Art)) +
  geom_point(size = 3, alpha = 0.7, color = "darkblue") +
  geom_hline(yintercept = 0, color = "red", linetype = "dashed") +
  geom_text(aes(label = ifelse(abs(Residuals) > 0.5, Art, "")),
            vjust = -0.5, hjust = 0.5, size = 3) +
  labs(
    x = "Angepasste Werte (log scale)",
    y = "Residuen",
    title = "Residuen vs. Angepasste Werte"
  ) +
  theme_minimal()
```

5.2. Statistische Outputs

Vollständige Regressionsanalyse

```
{r}
#| label: tbl-full-regression
#| tbl-cap: "Detaillierte Regressionsstatistiken"

model <- lm(log(brain) ~ log(body), data = Animals)
summary_full <- summary(model)

# Create comprehensive statistics table
regression_stats <- data.frame(
  Statistik = c("Multiple R-squared", "Adjusted R-squared", "F-statistic",
                "F p-value", "Residual standard error", "Degrees of freedom"),
  Wert = c(
    round(summary_full$r.squared, 4),
    round(summary_full$adj.r.squared, 4),
    round(summary_full$fstatistic[1], 2),
    ifelse(pf(summary_full$fstatistic[1], summary_full$fstatistic[2],
             summary_full$fstatistic[3], lower.tail = FALSE) < 0.001,
           "< 0.001",
           round(pf(summary_full$fstatistic[1], summary_full$fstatistic[2],
                    summary_full$fstatistic[3], lower.tail = FALSE), 4)),
    round(summary_full$sigma, 4),
    paste(summary_full$df[2], "on", summary_full$df[1], "DF")
  )
)

kable(regression_stats)
```

Koeffizienten mit Konfidenzintervallen

```
{r}
#| label: tbl-coefficients
#| tbl-cap: "Regressionskoeffizienten mit 95% Konfidenzintervallen"

coef_table <- data.frame(
  Parameter = c("Intercept (log a)", "Steigung (b)"),
  Schätzung = round(coef(model), 4),
  Standardfehler = round(summary_full$coefficients[,2], 4),
  t_Wert = round(summary_full$coefficients[,3], 3),
  p_Wert = ifelse(summary_full$coefficients[,4] < 0.001, "< 0.001",
                  round(summary_full$coefficients[,4], 4)),
  CI_2.5 = round(confint(model)[,1], 4),
  CI_97.5 = round(confint(model)[,2], 4)
)

kable(coef_table)
```

Modelldiagnostik

```
{r}
#| label: fig-diagnostics
#| fig-cap: "Diagnostische Plots für die Regressionsanalyse"
#| fig-height: 8

par(mfrow = c(2, 2))
plot(model, which = 1:4, caption = c("Residuen vs. Angepasste Werte",
                                     "Q-Q Plot der Residuen",
                                     "Scale-Location Plot",
                                     "Residuen vs. Leverage"))
par(mfrow = c(1, 1))
```